

## Class 25: Coordinate Systems

If  $\mathbf{B} = \{ \mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n \}$  is a basis for  $V$  and  $\mathbf{x}$  is any vector in  $V$ , how do you find the coordinate vector for  $\mathbf{x}$  relative to  $\mathbf{B}$ ?

The entries in the vector  $[\mathbf{x}]_{\mathbf{B}}$  are

If  $V$  is just  $\mathbf{R}^n$ , you can find these weights with the following steps.

**Example** Find the coordinate vector for  $\mathbf{x} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$  relative to the basis  $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$  for  $\mathbb{R}^2$

### Two types of questions:

1. Given  $\mathbf{x}$  in standard coordinates, how do you find the coordinate vector with respect to another basis  $\mathbf{B}$ ?
2. Given the coordinate vector  $[\mathbf{x}]_{\mathbf{B}}$ , how do you find the standard coordinates for  $\mathbf{x}$ ?

### The coordinate mapping:

Let  $\mathbf{B} = \{ \mathbf{b}_1, \dots, \mathbf{b}_n \}$  be a basis for  $V$  and  $\mathbf{x}$  be any vector in  $V$ . The mapping that sends  $\mathbf{x}$  to its coordinate vector  $[\mathbf{x}]_{\mathbf{B}}$  in  $\mathbf{R}^n$  is called the coordinate mapping.

**Theorem:** The coordinate mapping is a linear transformation from  $V$  to  $\mathbf{R}^n$ .

Proof:

Show  $T$  preserves scalar multiplication

Let  $r$  be any scalar. Show  $T(r \mathbf{u}) = r T(\mathbf{u})$ .

**Practice:** Are the polynomials  $\{1 - 2t^2 - 3t^3, t + t^3, 1 + 3t - 2t^3\}$  linearly independent?

Write the polynomials in  $\mathcal{P}^3$  as coordinate vectors in  $\mathbb{R}^4$  relative to the basis  $\{1, t, t^2, t^3\}$  for  $\mathcal{P}^3$ .